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Was there SAMBA in Columbia? Paleomagnetic evidence from 1790 Ma Avanavero mafic sills (northern Amazonian Craton)

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ABSTRACT

The Amazonian Craton represents an important piece in the Paleoproterozoic paleogeography of the Earth. This study presents paleomagnetic data obtained on well-dated (U–Pb) 1790 Ma mafic sills from the Avanavero magmatism in northern Amazonian Craton (Guiana Shield). AF and thermal treatments revealed southeastern, low downward/upward inclination, remanent magnetization directions that are carried by moderate to high-H_c and high-T_B Ti-poor titanomagnetite. The site mean directions cluster around the mean $D_m = 138.2°$, $I_m = -3.4°$ ($N = 13$, $\alpha_{95} = 13.0°$), which yields a robust paleomagnetic pole (AV pole) at 27.9°E, 48.4°S ($A_{95} = 9.6°$) with a Q-value of 5. The characteristic component disclosed for the Avanavero sill matches that obtained for sediments collected along the baked contact and are distinct from those away from the sill. The Avanavero directions are also significantly different from those obtained for younger units, with ages spanning from 1420 to 520 Ma, suggesting the sills carry a primary remanence. The Avanavero pole helps in constraining the paleogeography of the central pieces of Columbia. It is compatible with the Baltica and Amazonian Cratons SAMBA link in the Columbia Supercontinent at about 1780 Ma, but other configurations are also possible. When compared to other Paleo- to Mesoproterozoic paleomagnetic poles from the southern Amazonian Craton (Central Brazil Shield), our new paleomagnetic pole suggests intracratonic motions within the southern area of the Craton, probably after 1420 Ma ago. We tentatively suggest that these movements are related to the collision of the Paraguá Block with the proto-Amazonian Craton at about 1350–1320 Ma ago.

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1. Introduction

The paleogeography of Proterozoic supercontinents is still controversial because of the difficulties and uncertainties that arise when someone compares the geological and geophysical record of continental fragments that once have come together to form them (Pesonen et al., 2003). Nevertheless, it is believed that at least twice in the Proterozoic continents assembled to form supercontinents (Hossain et al., 2007): at ca. 1850 Ma ago with the Columbia Supercontinent (e.g., Rogers, 1996; Rogers and Santosh, 2002; Zhao et al., 2004, 2006, 2008; Meert, 2012; Kusky et al., 2007; Johansson, 2009; Yakubchuk, 2010), and at ca. 1100–1000 Ma with the Rodinia supercontinent (e.g., Hoffman, 1991; Dalziel, 1997; Weil et al., 1998; D'Agrella-Filho et al., 1998; Pesonen et al., 2003; Li et al., 2008; Evans, 2009).

The exact configuration of Columbia is still unknown and a great deal of geological, geochronological and paleomagnetic data is required in order to provide a means of establishing the links between the different continental fragments in the Paleo- and Mesoproterozoic. From the paleomagnetic point of view, only precisely dated Paleoproterozoic geological units of these continental fragments have the potential to provide reliable information to constrain the Columbia paleogeography (Buchan et al., 2000; Pesonen et al., 2003). Gradually, these data are being acquired in several continents and we can now propose some crude quantitative tests on the different Columbia configurations using the paleomagnetic database (e.g., Meert, 2002, 2012; Pesonen et al., 2003, 2012; Salminen and Pesonen, 2007; Bispo-Santos et al., 2008, 2012; Johansson, 2009; Cordani et al., 2009; Lubnina et al., 2010; Piper et al., 2011; Evans and Mitchell, 2011; Zhang et al., 2012; D'Agrella-Filho et al., 2012; Mertanen and Pesonen, 2012).

Baltica and Laurentia are the central pieces of Columbia. There are basically two models for the Laurentia-Baltica connection in this supercontinent: the first one juxtaposes western Baltica to the southeastern Greenland in Laurentia (Zhao et al., 2004; Hou

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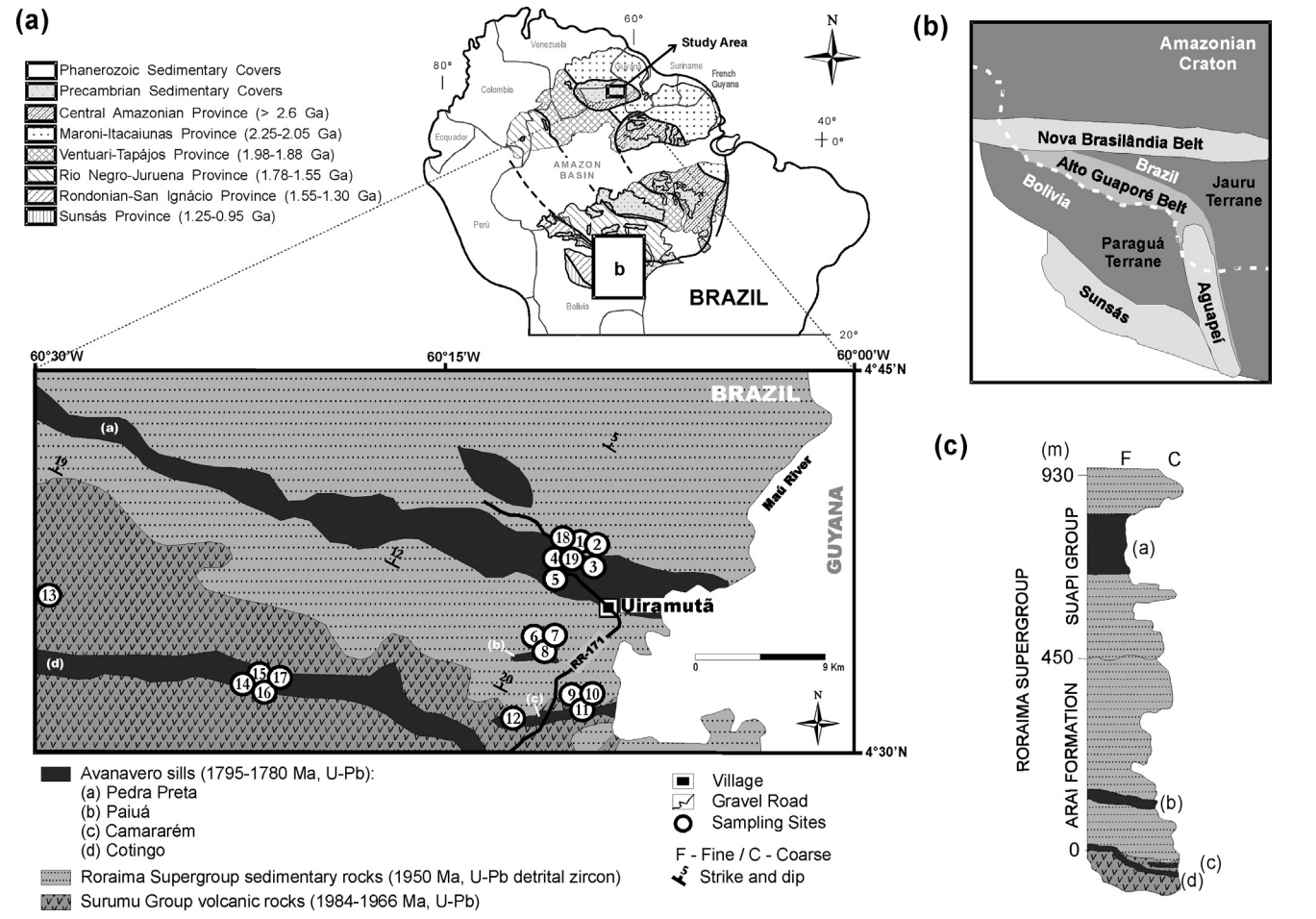


Fig. 1. (a) Geological sketch of the studied area with location of sampling sites (circled numbers) (adapted from Reis et al., 2004), inset: Amazonian Craton and their geochronological Provinces (after Tassinari et al., 2000); (b) general geometry of SW Amazonian Craton showing the ca. 1.47–1.35 Ga Alto Guaporé belt (Rizzotto and Hartmann, 2012), and the ca. 1.2–1.0 Ga Sunsás, Aguapeí and Nova Brasilândia belts (adapted from D'Agrella-Filho et al., 2012); (c) schematic stratigraphic section of Roraima Supergroup in the studied area (after Reis and Yánez, 2001).

et al., 2008; Johansson, 2009; Halls et al., 2011), the second one juxtaposes northern Baltica to eastern Greenland (Karlstrom et al., 2001; Buchan et al., 2000; Pesonen et al., 2003; Zhao et al., 2006; Salminen and Pesonen, 2007; Lubnina et al., 2010; Pisarevsky and Bylund, 2010; Evans and Mitchell, 2011; D'Agrella-Filho et al., 2012; Bispo-Santos et al., 2008, 2012). Paleomagnetic evidence favors the last link (Buchan et al., 2000; Salminen and Pesonen, 2007; Lubnina et al., 2010; Pisarevsky and Bylund, 2010). Also at the heart of Columbia, a link between northern South America (SAM) (Amazonian Craton) and southwestern Baltica (BA), dubbed SAMBA, was proposed on the basis of Archaean to Paleo-Mesoproterozoic geological evidence (e.g., Johansson, 2009). However, until recently the available Paleo- to Mesoproterozoic paleomagnetic data implied a gap between these two blocks (Pesonen et al., 2003; Bispo-Santos et al., 2008; D'Agrella-Filho et al., 2012). Based on 1790–1770 Ma paleomagnetic data from Baltica (Ropruchey sills and Shoksha Formation), Amazonian Craton (Colíder Group) and North China (Tahiang dykes), Bispo-Santos et al. (2008) suggested that North China could fill the gap between Baltica and the Amazonian Craton. Recent paleomagnetic data from 1420 Ma mafic dykes from Nova Guarita swarm and gabbros from the Indiavaí Intrusive (southern Amazonian Craton) seemed to corroborate this interpretation (D'Agrella-Filho et al., 2012). Here we present a paleomagnetic study of well-dated 1789 Ma mafic sills from the Avanavero magmatic event located at the northern Roraima State (Central Amazonian Province of the Amazonian Craton) and its implications for the paleogeography of Columbia.

2. Geological setting

The Amazonian Craton (Fig. 1a) is one of the largest stable cratonic areas in the world (430,000 km^2). It is exposed in two large areas separated by the Phanerozoic Amazonian sedimentary basin comprising the northern Guiana Shield and the southern Central-Brazil or Guaporé Shield (Lacerda-Filho et al., 2004; Schobbenhaus et al., 1984; Santos et al., 2000). Its northeastern part consists of two Archean nuclei (>2600 Ma – Central Amazonian Province in Fig. 1a) encircled by a predominantly accretionary Paleoproterozoic belt (2250–2050 Ma) which form the Maroni-Itacaiúnas Province (Tassinari and Macambira, 1999; Tassinari et al., 2000). The southwestern part of this Archean nucleus was accreted by subduction-related juvenile magmatic arcs forming the 1980–1810 Ma Ventuari-Tapajós, and the 1780–1600 Ma Rio Negro-Juruena Provinces (Tassinari et al., 2000; Pinho et al., 2003; Schobbenhaus and Brito-Neves, 2003; Cordani and Teixeira, 2007). In the Mesoproterozoic, subduction-related magmatic arcs of the 1600–1300 Ma Jauru Terrane in the Central-Brazil Shield (Rondonian-San Ignacio Province) developed until the final collision of the Paraguá Terrane at ca. 1320 Ma (Bettencourt et al., 2010). This collisional model was recently extended to the northwest with the recognition of the Trincheira ophiolite by Rizzotto and Hartmann (2012). These authors interpret these rocks as a record of oceanic crust relics associated with the collision of the Paraguá Block with the proto-Amazonian Craton along the Alto Guaporé Belt at the Middle Mesoproterozoic (Fig. 1b). In this geotectonic

scenario the E-W 1100–1000 Ma Nova Brasilândia belt (NBB) located between the proto-Amazonian Craton and the Paraguá block, and once interpreted as a suture zone between these two cratonic blocks (Tohver et al., 2004), most probably represents intracratonic reactivations occurred during the development of the Greenvile-type 1200–950 Ma Sunsás-Aguapeí Belt (e.g., Teixeira et al., 2010). Similar 1420 Ma paleomagnetic poles obtained for the Nova Guarita dykes and the Indiavaí intrusive, whose areas are located on opposite sides of the NBB, corroborate this intracratonic interpretation (D'Agrella-Filho et al., 2012).

The study area is located in the Central Amazonian Province (CAP; Fig. 1a), which is composed of Archaean continental crust (>2600 Ma) affected by significant Paleoproterozoic magmatic events. The study area is part of a CAP domain known as Roraima Block. This block is completely covered by the 1980–1960 Ma Surumu acid volcanic rocks and the overlying ca. 1870 Ma sedimentary sequences of the Roraima Supergroup, that includes the basal Arai Formation, the Suapi Group and the Uaimapué Formation at the top (Fig. 1c) (Teixeira et al., 1989; Tassinari and Macambira, 1999; Santos et al., 2003). The Roraima basin is possibly related to a N-S extensional event, responsible for the development of normal E-W faults and transference faults. It has been interpreted as an intracontinental type basin with regional rhombohedral geometry controlled by NW-SE system faults. Later, during the K'Mudku Episode around 1.2 Ga the basin experienced a weak positive inversion involving the reactivation of normal upthrow faults, where the transfer faults worked as lateral ramps. At the southern border of the basin along the Brazilian side, in the proximity of the E-W faults, the Roraima Supergroup was affected by *en echelon* folds. Towards the center of the basin, no deformation was recorded in these rock units, suggesting their primary stratigraphic relationships were preserved (Reis et al., 2012).

Mafic sills and dykes of the Avanavero magmatic event intruded into the volcanic and sedimentary formations of the Surumu Group and the Roraima Supergroup (Brito Neves, 2011; Reis et al., 2010, 2013). The Avanavero magmatism extends to the surrounding countries of Venezuela, Guyana and Suriname, and forms a large igneous province (LIP). The sills follow the original structures of the Roraima sedimentary rocks, i.e., their bedding planes. In the study area these structural planes have northwest-southeast strikes and dip to northeast between 5° and 20° (Fig. 1c). Groeneweg and Bosma (1969) were the first to use the name Avanavero to designate dolerite dykes and large irregular sheet-shaped bodies in the Avanavero area and other locations in Suriname. In Brazil, this Proterozoic basic magmatism comprises four levels of sills that are informally called Cotingo, Camararém, Paiuá and Pedra Preta sills (Pinheiro et al., 1990). The Cotingo sill intrudes the Surumu volcanic rocks. It is about 140 km long and 40 m thick and strikes WNW-ESE. A small sill near the contact between the Surumu volcanics and the Arai sandstones is known as Camararém sill. It extends northwards to the Guyanas where it is known as Tumatumari sill. Sill Camararém is 30 meters thick and follows the orientation of the host sedimentary rocks dipping 20° to northeast (see Fig. 1c). Paiuá sill is 20 m thick. It cuts quartz sandstones at the first 200 meters of the Arai Formation and dips ca. 15° to the northeast (see Fig. 1c). Finally, the Pedra Preta sill cuts across the Suapi Group, where siltstones near the contact with the mafic rock were hornfelsed (Pinheiro et al., 1990). A thickness of about 250 meters is estimated for this sill that shows a gentle dip of around 12° to the northeast (Fig. 1c). Basic dykes associated with major faults (NE and NW trends) in the Roraima Supergroup are interpreted as the feeders of the poorly differentiated Avanavero dolerite sills. Mesozoic basic dykes associated with major mainly NE faults cut all the Proterozoic rocks (Reis et al., 2004).

The Avanavero rocks comprise dark-gray medium to coarse grained dolerites and gabbros, and subordinately by microdiorites and micro-quartz diorites (Reis et al., 2013). Under the microscope they are composed of plagioclase, pyroxene and amphibole, with magnetite and ilmenite as accessory minerals. Late hydrothermal processes are common, such as alteration of plagioclase to clay minerals and/or albite (Pinheiro et al., 1990). Dolerite dykes show generally subophitic texture, and are composed of plagioclase (labradorite), pyroxene (augite and occasionally pigeonite), hornblende and chlorite. Sericite and epidote appear as accessory minerals (Reis et al., 2013).

U–Pb ages in baddeleyite were obtained for several dolerite dykes and sills of the Avanavero magmatic suite. These comprise ages of 1794 ± 4 Ma for a dolerite intrusive into the Omai gold deposit in Guyana (Norcross et al., 2000), a 1787 ± 14 Ma age for the Cipó dolerite and a 1782 ± 3 Ma age for the Cotingo dolerite (Santos et al., 2003). Similar ages were also obtained for correlative mafic intrusions in Pará and Amazonas Brazilian States, located several hundreds of kilometers south of the studied area. They comprise a 1780 ± 7 Ma age for the Crepori sill in southwest Pará State, and a 1780 ± 3 Ma age for the Quarenta Ilhas sill in northeast Amazonas State (Santos et al., 2002). Recently, Reis et al. (2013) presented two new U–Pb baddeleyite ages for Avanavero rocks: a 1795 ± 2 Ma age for the Pedra Preta sill, and a 1793 ± 2 Ma age for a dolerite sill cross-cutting the Urupi Formation (Central-Brazil Shield), whose rocks have been correlated with those from the Roraima Supergroup. A mean age of 1788.5 ± 2.5 Ma was calculated from all U–Pb ages and is assumed to be the best age estimate for the Avanavero magmatic event. Besides, ^{40}Ar–^{39}Ar analysis on biotite of a dolerite located in Guyana yielded an integrated age of 1798 ± 2 Ma (Onstott et al., 1984) that is in agreement with the U–Pb ages dataset. This age and the fact that the Arai Group show only anchimetamorphic mineral assemblages are strong evidence that the region was not submitted to temperatures higher than 300 °C after the emplacement of the Avanavero sills.

3. Sampling and methods

Paleomagnetic sampling was performed around the small village of Uiramutã at the northern Roraima State (Brazil) where 118 standard oriented cylindrical cores were taken from 19 sites of four well-exposed sills (Cotingo, Camararém, Paiuá and Pedra Preta) from the Avanavero suite (Fig. 1, Table 1). From the Pedra Preta sill, estimated to be 250 m thick, we sampled 7 sites (#1 to #5, #18, #19 in Table 1 and Fig. 1) at different levels of the sill. Three sites (#6, #7 and #8) were sampled from the Paiuá sill, estimated to be 20 m thick. Four sites were sampled from the Camararém sill, estimated to be 30 m thick, and five sites (#13 to #17) from the ca. 40 m thick Cotingo sill. Samples were oriented using both sun and magnetic compasses. A baked contact test was attempted at site #6 at the border of the Paiuá sill with host sandstones (Fig. 2). At this point, the Paiuá sill is covered by six meters of sandstones dipping 15° to the northeast. In addition to the 8 cylinders collected in the sill, we have sampled six oriented cores from the sediments at the following distances from the contact: 300 cm (FR11-I), 326 cm (FR11-J), 366 cm (FR11-K), and 516 cm (FR11-L, FR11-M, FR11-N) (Fig. 2).

At the laboratory, cylindrical cores were cut into 2.2 cm height specimens. Step-wise alternating magnetic field (AF) and thermal demagnetization techniques were employed to separate the characteristic remanent magnetization (ChRM) component carried by the samples. Steps of 2.5 mT (up to 15 mT) and 5 mT (15–100 mT) were employed for AF demagnetization using a 2-axis tumbler Molspin AF demagnetizer. Thermal demagnetization was performed using a Magnetic Measurements TD-60 furnace in steps of 50 °C (from 150 °C up to 500 °C) and 20 °C (from 500 °C up to 600 °C). Remanent magnetization measurements were carried out using

Table 1
Paleomagnetic data from Avanavero mafic sills.

Site	Localization	Samples	Site mean direction					VGP	
			N/n	Dec (°)	Inc (°)	α_{95} (°)	K	Plong (°E)	Plat (°N)
1	04°37.8′N/60°09.6′W	FR1	15/15	140.6	12.5	4.0	92.9	14.6	−49.2
2	04°37.8′N/60°09.6′W	FR2	16/16	139.8	12.4	3.7	102.2	15.0	−48.4
3	04°37.2′N/60°09.6′W	FR3	13/12	135.8	12.7	4.6	89.8	16.1	−44.5
4	04°37.2′N/60°10.2′W	FR4	12/12	140.2	−6.9	3.4	164.7	29.7	−50.3
5	04°36.6′N/60°10.2′W	FR5	17/12	103.4	−13.9	2.8	249.2	36.0	−13.8
6	04°34.2′N/60°11.4′W	FR11	18/13	151.4	26.6	4.7	79.0	3.9	−56.1
7[a]	04°34.2′N/60°11.4′W	FR12[a]	11/5	351.4	28.1	11.8	42.7	–	–
8[a]	04°34.2′N/60°11.4′W	FR13[a]	13/5	9.2	34.9	10.2	56.9	–	–
9	04°31.8′N/60°09.6′W	FR85	18/12	128.9	−17.2	6.6	44.6	37.5	−39.1
10[a]	04°31.8′N/60°09.6′W	FR86[a]	14/14	348.8	−33.7	4.8	69.1	–	–
11[a]	04°31.8′N/60°09.6′W	FR87[a]	10/0	–	–	–	–	–	–
12[a]	04°31.2′N/60°12.6′W	FR88[a]	14/13	8.8	2.8	4.1	103.5	–	–
13[a]	04°36.0′N/60°30.6′W	FR92[a]	10/0	–	–	–	–	–	–
14	04°32.4′N/60°19.8′W	FR143	12/12	157.2	−37.3	3.3	174.5	68.0	−62.5
15	04°32.4′N/60°19.8′W	FR144	14/14	147.0	−21.7	2.1	366.2	43.3	−56.7
16	04°32.4′N/60°19.8′W	FR145	13/5	115.8	11.3	8.8	76.9	21.2	−25.1
17	04°32.4′N/60°19.8′W	DB328-330	14/11	162.7	−21.8	8.8	28.0	52.1	−71.6
18	04°37.8′N/60°09.8′W	DB342-343	10/9	136.8	12.7	4.0	164.6	15.8	−45.5
19	04°37.1′N/60°09.9′W	DB344-345	10/10	141.2	−12.4	3.2	223.0	34.1	−51.3
Mean			**19/13**	**138.2**	**−3.4**	**13.0**	**11.1**		
AV pole						**9.6**	**19.8**	**27.9**	**−48.4**

n/N – number of analyzed samples/number of samples used in the mean; Dec. – declination; Inc. – inclination; α_{95}, K – Fisher's confidence cone and precision parameter (Fisher, 1953); VPG – Virtual Geomagnetic Pole; Plong – Paleolongitude; Plat – Paleolatitude.
[a] Excluded sites from the mean.

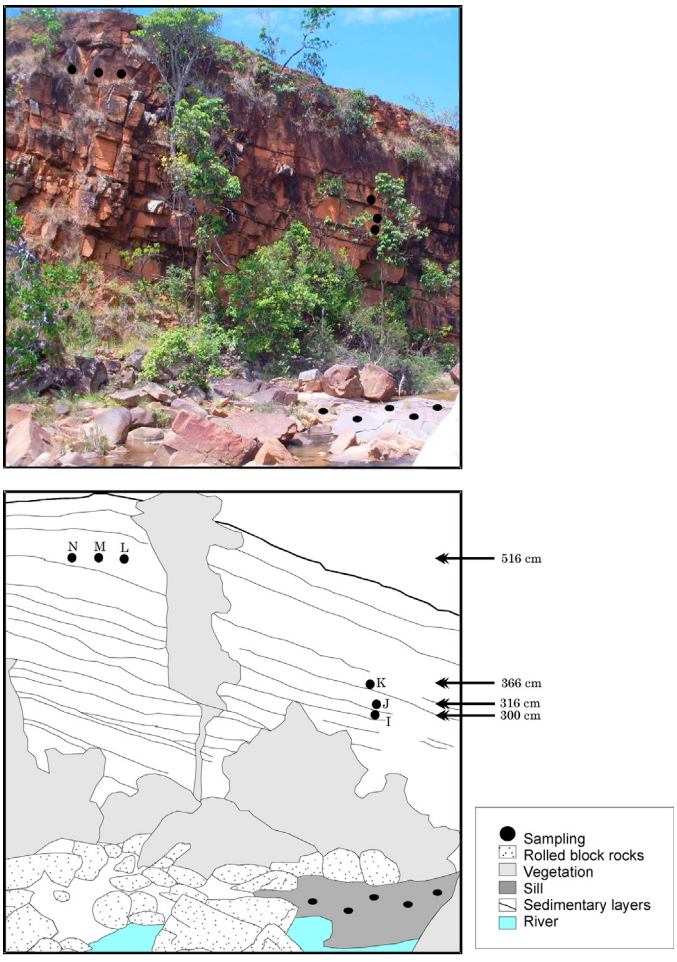


Fig. 2. Photo and schematic representation of the sampled site for the baked contact test showing distances between the mafic sill and the host sandstone and location of samples.

a JR-6A spinner magnetometer (AGICO, Czech Republic). These instruments are housed in a magnetically-shielded room with ambient field <1000 nT. Magnetic components for each specimen were identified in orthogonal plots (Zijderveld, 1967), and calculated using the principle component analysis (Kirschvink, 1980). At least 4 demagnetization steps were used to calculate vectors using least-squares fits, and an upper limit for mean angular deviation (MAD) of 8° was used. Fisher's (1953) statistics was used to calculate site mean directions and the paleomagnetic pole.

Magnetic mineralogy was investigated through acquisition of isothermal remanent magnetization (IRM) in a pulse magnetizer (Magnetic Measurements), and hysteresis curves using a MicroMag VSM – Model 3900 (Princeton Measurements Corporation). These curves give bulk coercive force (H_c), saturation magnetization (M_s), and saturation isothermal remanence (M_{rs}), after subtraction of the paramagnetic contribution from the high field portion of the curve. Coercivity of remanence (H_{cr}) was obtained using the same Micromag VSM instrument. Also, thermomagnetic curves (low-field magnetic susceptibility versus (high and low) temperature) were performed for several samples to characterize the magnetic carriers in the rock, using a CS-4 apparatus coupled to the KLY-4S Kappabridge instrument (AGICO, Czech Republic). Thin and polished sections were microscopically investigated under transmitted and reflected light, respectively.

4. Paleomagnetic results

NRM intensities vary between 0.975 A/m and 459 A/m, which are typical values for basaltic rocks, with high contribution of ferrimagnetic minerals. Stable characteristic remanent magnetization (ChRM) directions were isolated after both AF and thermal demagnetizations of 153 samples from 13 out of the 19 analyzed sites. Scattered secondary components were removed in fields up to 20–30 mT, and in temperatures up to 540 °C, after which southeastern directions with low to moderate upward/downward inclinations were disclosed (Fig. 3).

As for the remaining sites, three of them (sites #7, #8 and #12 in Table 1) presented a northern direction with low to moderate downward inclination. This component is similar to that found for

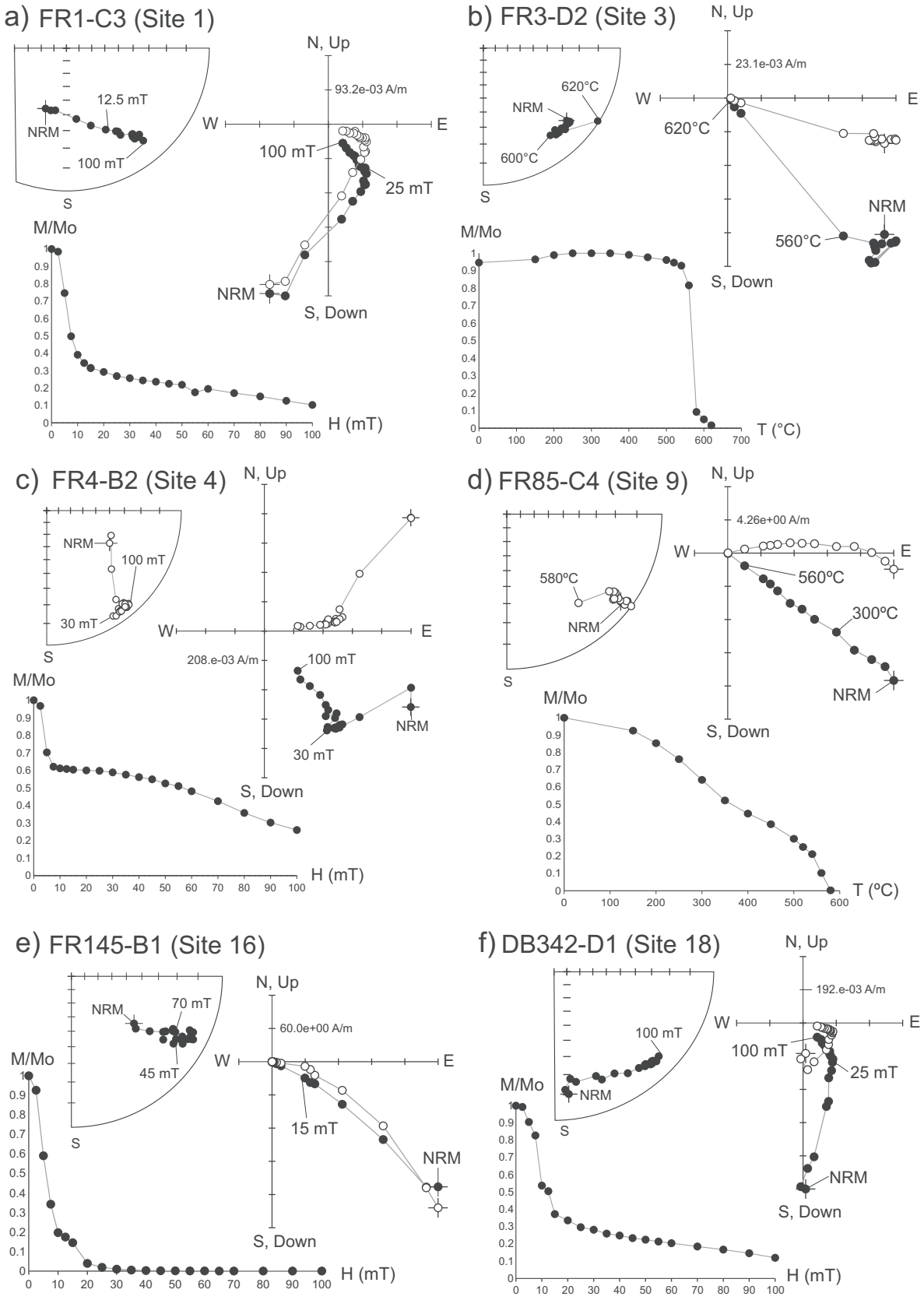


Fig. 3. Examples of magnetic component obtained for samples from six sites after AF and thermal demagnetizations. Figure shows stereographic projections (solid (open) symbols represent positive (negative) inclinations), normalized magnetization intensity curves (M/M_o versus alternating field (H)) and orthogonal projections (solid (open) symbols represent horizontal (vertical) projections) for each sample.

Jurassic (~200 Ma) dolerites collected in Suriname (South America) by Veldkamp et al. (1971) suggesting these rocks are younger or were remagnetized by this Permo-Triassic magmatic event. Samples from site #10 yielded northwestern directions with upward inclination not far from the present-day geomagnetic field. Samples from sites #11 and #13 yielded unstable behavior after AF and thermal treatment. Results from these 6 sites were excluded from the mean directions and paleomagnetic pole determinations.

Table 1 presents a summary of paleomagnetic results. The 13 Avanavero site mean ChRM directions (Fig. 4) cluster around the mean, $D_m = 138.2^\circ$, $I_m = -3.4^\circ$ ($\alpha_{95} = 13.0^\circ$, $K = 11.1$), which yielded the paleomagnetic pole located at 27.9°E, 48.4°S ($A_{95} = 9.6^\circ$, $K = 19.8$) (AV pole). The calculated angular dispersion ($s = 18.2^\circ$) is compatible with an adequate sampling of geomagnetic secular variation. After tilt-correction the direction does not change significantly, with a mean of $D_m = 139.5^\circ$, $I_m = -2.0^\circ$ ($\alpha_{95} = 13.0^\circ$, $K = 11.2$).

5. Magnetic mineralogy

Under alternating magnetic field (AF) treatment all samples presented mean destructive fields of less than 20 mT, but two distinct coercivity distributions can be observed: (i) for some samples the intensity of magnetization falls to less than 5% of the NRM before 20 mT demagnetization (samples FR11-C2 and FR145-A4 in Fig. 5a), whereas (ii) other samples preserve more than ~10% of the total intensity at fields as high as 100 mT (samples FR4-F2 and FR1-D3 in Fig. 5a). The low-coercivity magnetic grains are usually associated with secondary components, while moderate and high coercivity magnetic grains represent the ChRM isolated in the samples. Thermal demagnetization showed two types of unblocking temperature spectra: (i) a more common discrete spectrum, where unblocking temperatures are limited to the range between 500–550 °C and 600 °C (samples FR2-B5, FR3-D2 and FR88-E1 in Fig. 5b) and (ii) a distributed spectrum, where it is possible to distinguish two wide ranges of unblocking temperatures between 0 °C and 450–500 °C and between 500 °C and 600 °C (sample FR85-C4 in Fig. 5b). The characteristic remanent magnetization (ChRM) is normally associated with the highest unblocking temperatures, whose magnetic carriers are represented by Ti-poor titanomagnetite, as shown by the decay intensity curves.

Isothermal remanent magnetization (IRM) acquisition curves for all samples are very similar, giving saturation fields of less than 300–400 mT (Fig. 5c). These coercivities are typical of magnetite or titanomagnetite minerals. Narrow-waisted hysteresis curves were obtained for samples from all sites, giving coercivities between 4.3 mT and 24.9 mT, also typical of (titano)magnetite grains. In the Day's plot (Day et al., 1977, modified after Dunlop, 2002) most samples fall into the pseudo-single domain (PSD) field, or along a trend parallel to the theoretical SD/MD mixing curves of Dunlop (2002) (Fig. 5d). This is consistent with the stable ChRM directions disclosed by thermal and AF treatment.

Both reversible and irreversible high-temperature (20–700 °C) thermomagnetic curves were obtained, most of them with a pronounced Hopkinson peak and a strong decay in susceptibility at around ~580 °C (Fig. 5e and f). Low-temperature thermomagnetic curves show a susceptibility peak at around −150 °C denoting the Verwey transition. These characteristics are typical of low-Ti titanomagnetite grains (Dunlop and Ozdemir, 1997), which are the main magnetic carriers of ChRM direction in the studied rocks.

6. Petrography

Petrographic analyses were performed in samples with consistent ChRM directions and also in samples showing unstable and/or inconsistent directions. Mineralogical compositions and textures observed in thin and polished sections for samples with different magnetic behavior typically show differences that may explain their contrasting paleomagnetic results.

Samples with stable magnetization show subophitic textures, consisting predominantly of plagioclase (ca. 50–55%), augite (35–40%), biotite (2–5%), opaque minerals (2–5%), hornblende (< 1%) and epidote (< 1%). Some samples (FR4-F1, FR5-G1, FR85-B1, FR144-C3, and FR145-A1) show hornblende rims replacing augite, and occasionally myrmekite (samples FR1-B1 and FR4-F1). In polished section, the preserved euhedral magnetite crystals (Fig. 6A and B) are interpreted as the carriers of the ChRM. Minor quantities of hematite and martite (Fig. 6C) are present and probably result from late magmatic oxidation, but do not contribute significantly as remanence carriers of the ChRM. Occasionally, anhedral crystals of magnetite or hematite are observed as intergranular growths. Locally, magnetite grains present fine ilmenite lamellae, diagnostic of high-temperature oxidation-exsolution processes (Fig. 6B). Magnetite is rarely observed in thin-sections of samples FR143-A1, FR144-C3, and FR145-A1, although it is the main magnetic carrier of the ChRM direction.

Samples with unstable magnetization consist predominantly of plagioclase (ca. 50–55%), hornblende (10–20%), augite (0–20%), quartz (0–10%), biotite (5–10%), and opaque minerals (5–10%). Thin and polished sections indicate features typical of hydrothermal processes, such as hornblende appearing as alteration rims in augite crystals or in opaque minerals such as hematite (Fig. 6D). Moreover, a higher percentage of myrmekitic textures is observed in these samples. Opaque mineral are predominantly hematite, martite, and locally preserved magnetite (Fig. 6E). Corrosion features (Fig. 6D) are frequent in these samples, which could be the result of dissolution of (or substitution by) new minerals (hornblende or biotite). Martite crystals present patches or venules of thick ilmenite (Fig. 6F).

In addition to the mafic intrusions, we have also examined thin-sections from the host sandstones close and away of the baked contact in site #6. Close to the contact the sandstones show magnetite crystals with skeleton forms (Fig. 7A), sometimes partially replaced by martite (Fig. 7B). Away from the contact, martite replacing magnetite grains and hematite as cement are more frequent (Fig. 7C and D).

7. Baked contact test

Fig. 8 shows the demagnetization behavior of samples collected at the Paiuá mafic sill and its host sandstones. Mean directions for the sill ($D_m = 151.4^\circ$; $I_m = 26.6^\circ$, $\alpha_{95} = 4.7^\circ$) and the sandstones away from the contact ($D_m = 3.6^\circ$; $I_m = 12.9^\circ$, $\alpha_{95} = 20.4^\circ$) are indicated in Fig. 9. Most sandstone samples collected up to 366 cm away from the contact (FR11-I, FR11-J and FR11-K) showed a direction similar to that found in the intrusion (Fig. 9a). In contrast, sandstone samples (FR11-L to FR11-N) collected 516 cm away from the sill's border showed a consistent north-northwestern magnetic component with low inclination (Figs. 8 and 9). The age of the characteristic component of the sandstones away from the contact is still unconstrained. These directions are close to the magnetic components found in Jurassic dolerite dykes of the Guiana Shield (Veldkamp et al., 1971; Théveniaut et al., 2006), and may thus represent a younger remagnetization of the sandstones. Some sites on the mafic sills collected in the present study also provided similar directions (Table 1). But most sites in the sills were not affected by this thermal event. The fact that Ar–Ar dates on biotite are similar to the crystallization ages of the sills also suggest that the region has never been heated above 300 °C after the Avanavero magmatism. So, any remagnetization must be of chemical origin, related to

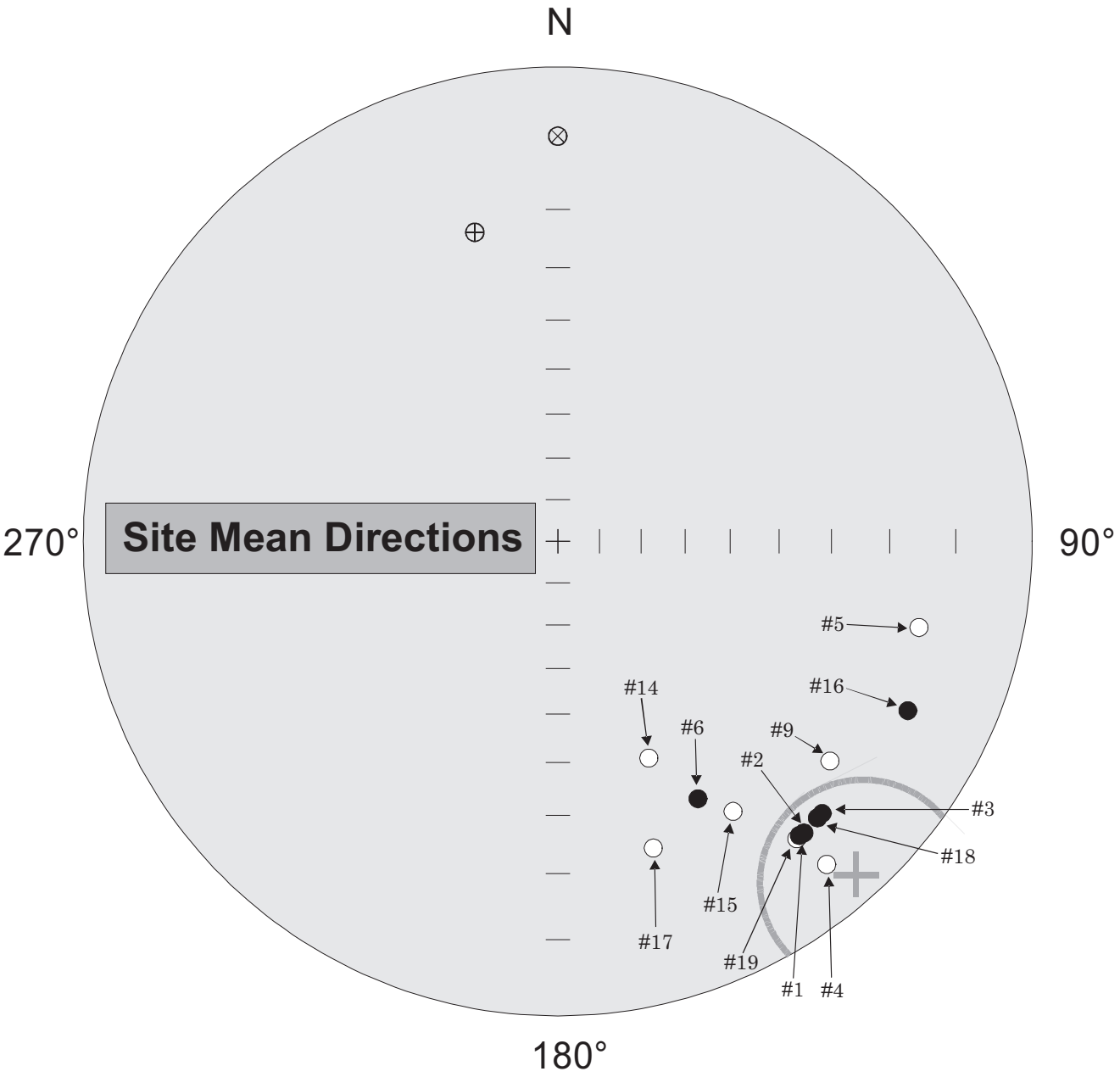


Fig. 4. Site mean directions (small solid and open circles) for the Avanavero rocks. Plus symbol represents the mean of site mean directions with its respective confidence circle ($\alpha_{95} = 13.0^{\circ}$). Solid (open) symbols represent downward (upward) inclinations. ⊗ – present dipolar geomagnetic field; ⊕ – present geomagnetic field. Numbers (#) indicate the analyzed sites.

fluid circulation at the time of the Jurassic Central Atlantic Province magmatism. This younger event likely affected the magnetic mineralogy of the sandstones away of the contact, as indicated by the higher degree of martitization and hematitization observed in polished sections (Fig. 7C and D). IRM acquisition curves for samples close and far from the sill confirm these findings, with those close to the contact showing a normally smaller contribution of hematite when compared with those away from the intrusion (Fig. 9b). Hysteresis loops for the sediments are wasp-waisted as a result of the mixture between high- and low-coercivity phases (Fig. 9c and d). Coercivities for samples close to the contact vary between 6.5 and 7.9 mT, whereas those collected away from the contact show higher coercivities between 7.2 and 30.9 mT. In any case, if a Jurassic event has affected the porous sediments away from the baked contact, it seems to have spared the remanence of the sill and sediments close to its contact, suggesting that both the magmatic rock and the nearby sedimentary host kept the magnetic record of the time of the intrusion.

8. Discussion

8.1. The Avanavero paleomagnetic pole

Stable southeastern directions with low downward and upward inclinations (AV component) were disclosed in samples from the Avanavero sills. Magnetic experiments indicate that pseudo-single domain magnetite grains with moderate to high coercivities and high unblocking temperatures carry the AV component, attesting to its high magnetic stability through time. Although the baked contact test cannot be considered positive, since the primary

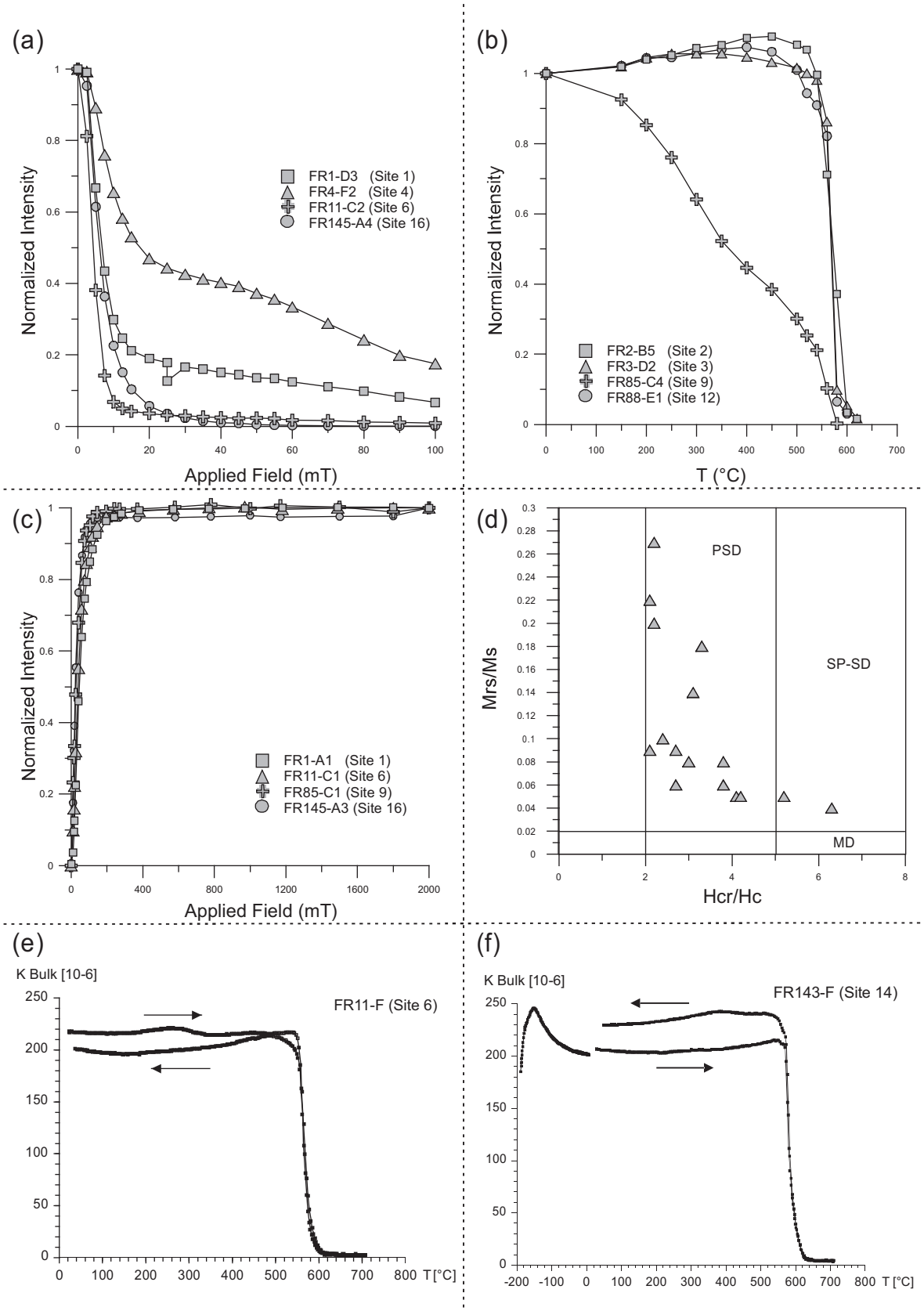


Fig. 5. Examples of normalized magnetization intensity curves: M/M_o versus alternating magnetic field (a) and temperature (b); Isothermal remanent magnetization (IRM) acquisition curves: normalized intensities versus magnetic field (c); Day's plot (Day et al., 1977) after Dunlop (2002) (d) – most samples fall in the Pseudo-single domain (PSD field) or along the SD (single domain) plus MD (Multi domain) mixing curves as proposed by Dunlop (2002); thermomagnetic curves showing variation in magnetic susceptibility K(SI) versus high and low temperature (e, f) (curves were corrected from furnace effects).

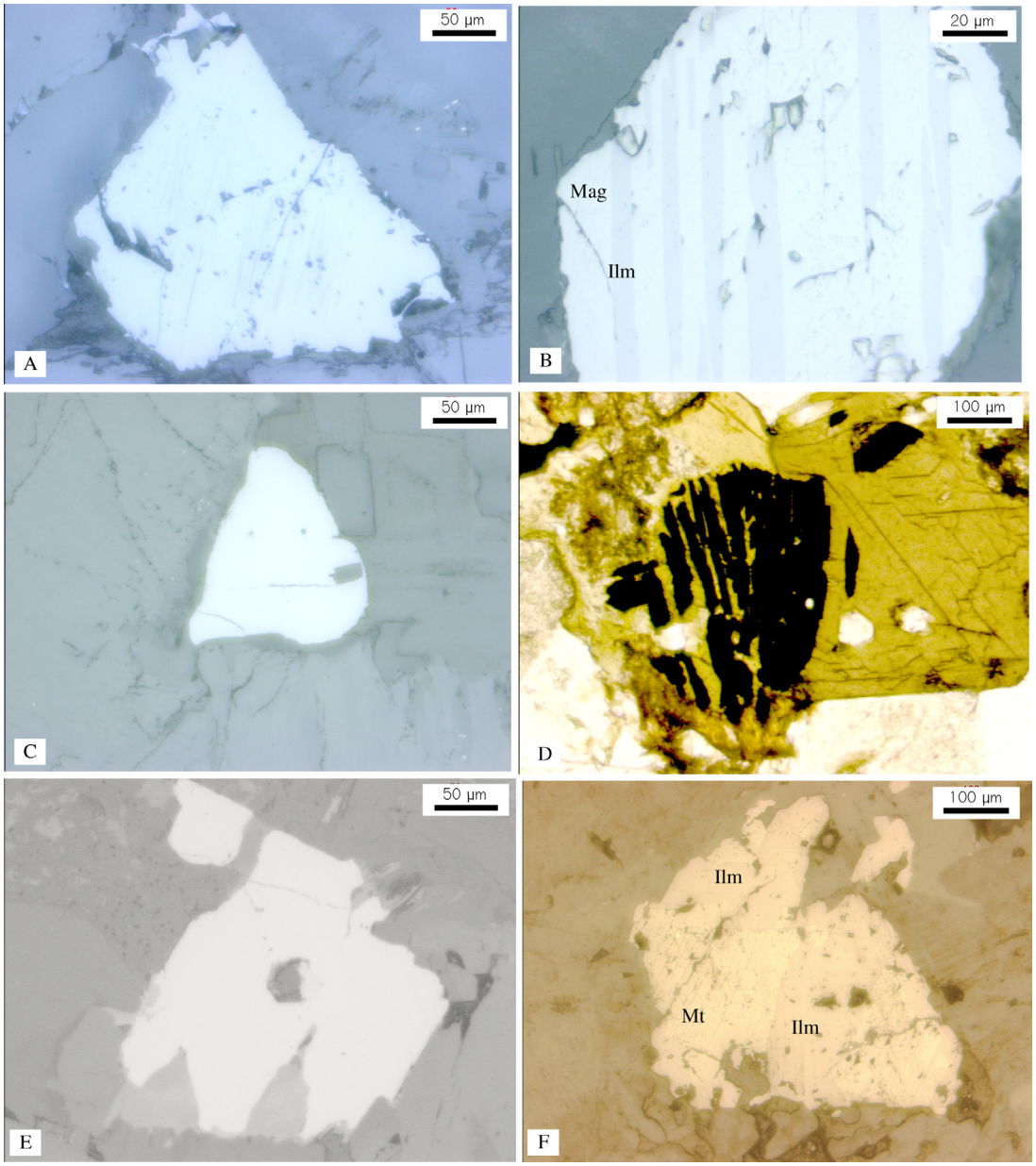


Fig. 6. Photomicrograph of analyzed samples from mafic sill: (A) Preserved euhedral magnetite crystal, (B) Magnetite grain with fine ilmenite lamellae, (C) Martite crystal, (D) Hematite crystal surrounded by hornblende, (E) Preserved Magnetite and (F) Martite crystal (bright color) patches of ilmenite. Polished sections with parallel polarizers.

nature of the magnetic component found for the host sandstones cannot be proved, the magnetic mineralogy, petrography and thermochronology described above together with the fact that the Avanavero pole does not match younger poles for the region, suggest the Avanavero ChRM direction likely represents a thermoremanent magnetization acquired during the cooling of mafic sills.

Since only four sills were sampled a question arises if secular variation was eliminated. The angular dispersion ($s = 18.2^\circ$) obtained in the Avanavero collection is compatible with an adequate sampling of geomagnetic secular variation. However, this dispersion may be related to later tilting that affected the sills. The intracratonic ca. 1200 Ma K'Mudku tectonic event affected the area to the SE of the Avanavero event (Cordani et al., 2010). It is considered to have been developed under low (to moderate) temperature conditions, and follows pre-existing E-W to NE-SW structural features that affected the country rocks at Paleoproterozoic times (Fraga et al., 2009). In spite of its regional importance in central Amazonia, this event did not affect the studied sills. As described above, the structural planes of the Roraima sedimentary rocks in the study area dip gently to the northeast, with inclinations increasing from northeast (dip of 5°) to southwest (dip of about 20°). Using the related structural dips at each collected sill, bedding plane corrections for each related sill were performed. A new site mean direction was calculated ($D_m = 139.5^\circ$; $I_m = -2.0^\circ$; $\alpha_{95} = 13.0^\circ$; $K = 11.2$), which differs by only one degree in declination and inclination from the mean direction obtained before bedding correction. The Avanavero ChRM directions are similar to those found for the Aro and Guaniamo dykes from Venezuela, dated at 1820 ± 20 Ma (Onstott et al., 1984), which are located ca. 600 km away from our sampling area. The similarity in the obtained poles suggests that no significant tilting affected the region.

If the sills were not greatly affected by tilting as shown above, the only possible way that site mean directions express the record of secular variation is to assume relatively slow cooling rates for the sills. The cooling rate of magmatic rocks depends of various factors, such as the thickness of the intrusion, its composition, and

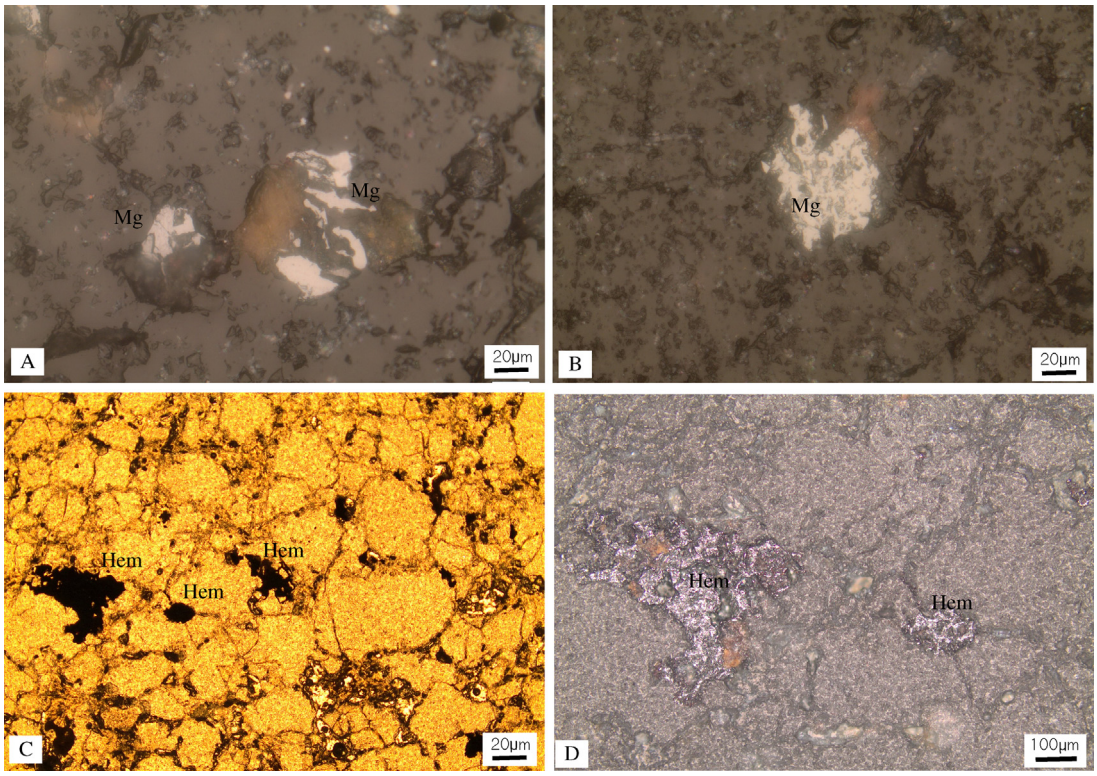


Fig. 7. Photomicrograph of samples from sedimentary rocks collected for the baked contact test: (A and B) Magnetite partially martitized (Polished sections with parallel polarizers), (C and D) hematite cementation in quartz-rich sandstone (Polished sections with transmitted light (C) and reflected light with parallel polarizers (D), respectively).

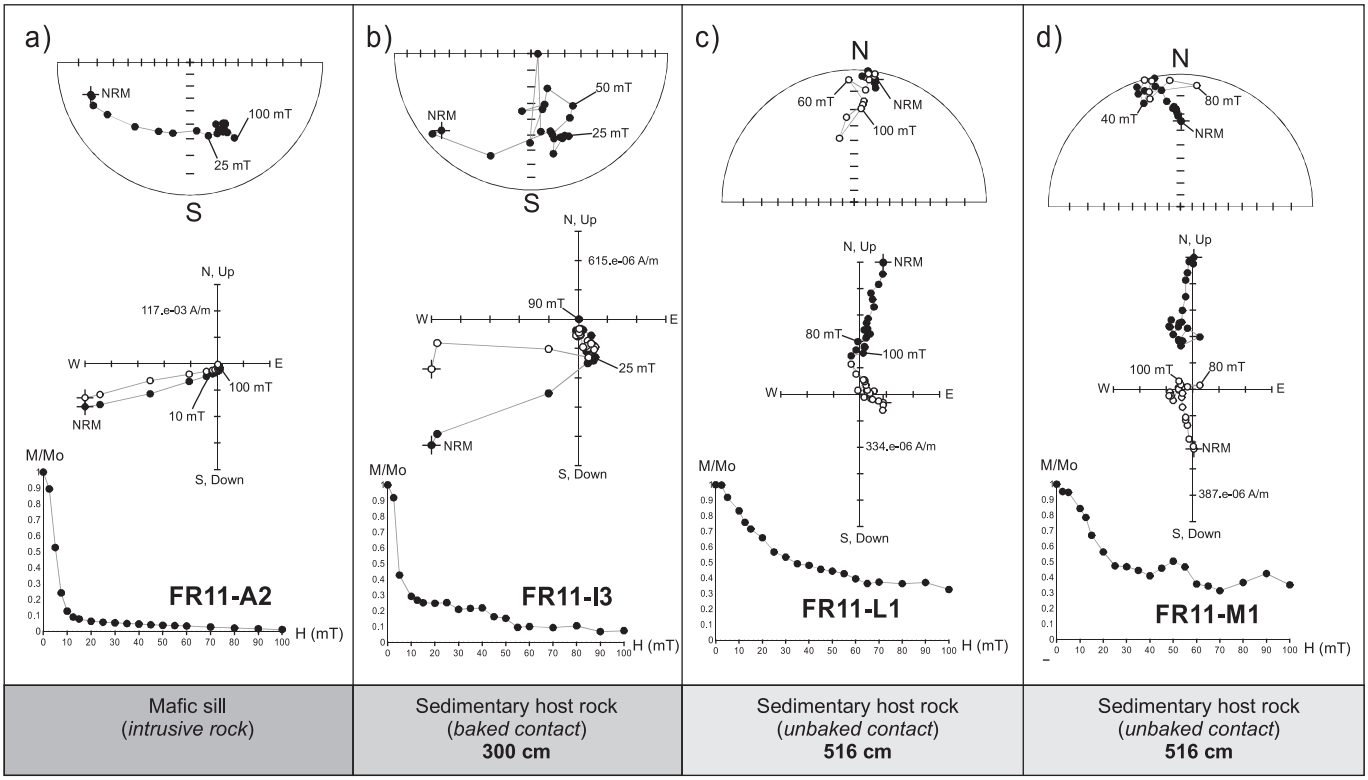


Fig. 8. Baked contact test: AF demagnetization of a sample from the mafic sill (a), of a baked contact sedimentary sample at 300 cm from intrusion (b), and of the sedimentary rocks (unbaked contact) at 516 cm (c) and (d). The figure shows stereographic projections (solid (open) symbols represent positive (negative) inclinations), orthogonal projections (solid (open) symbols represent horizontal (vertical) projections), and normalized magnetization intensity curves (M/M_o versus alternating field (H)) for each sample.

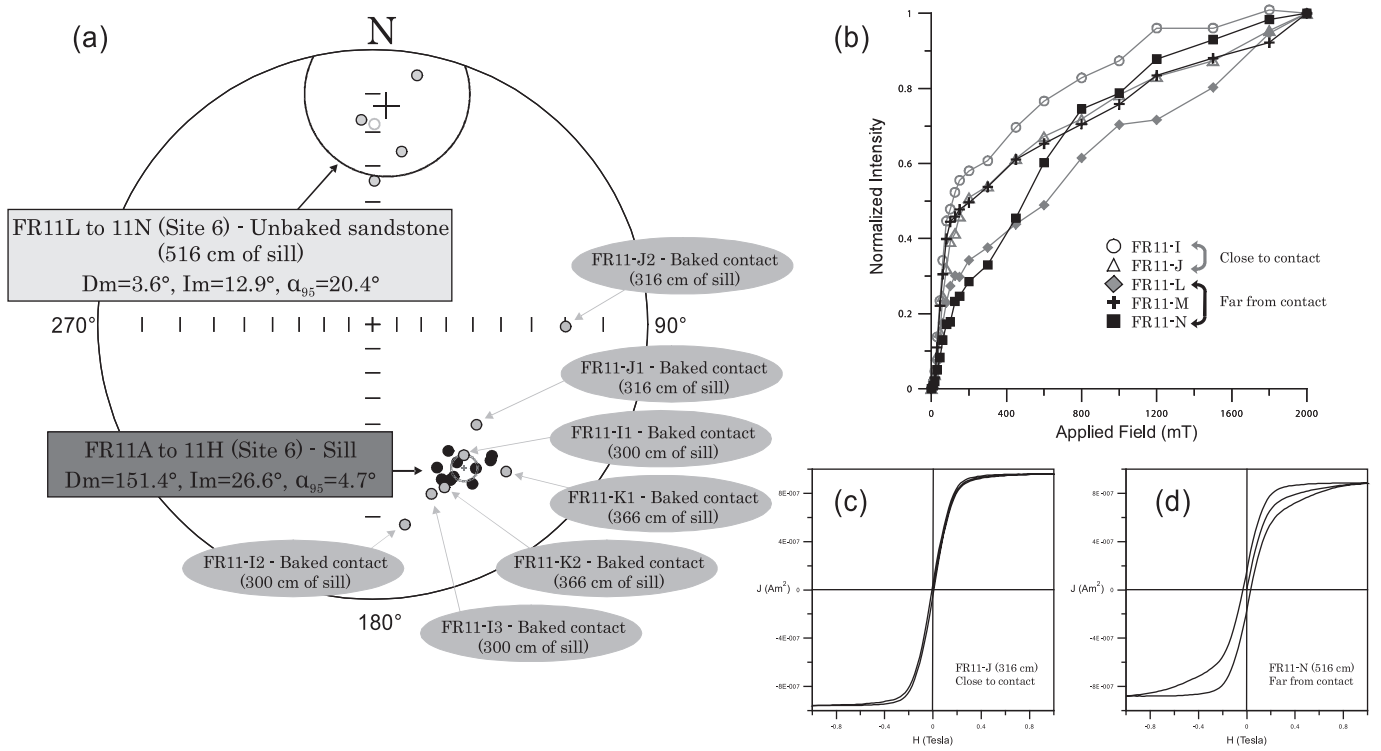


Fig. 9. (a) Sample magnetization directions for the mafic sill (FR11-site #6) and host sedimentary rocks. Solid/open symbols represent positive/negative inclinations. Site mean direction (plus symbol) and respective confidence circle are also shown for the sill (Dec = 151.4°, Inc = 26.6°, α_{95} = 4.7°) and for the host sandstone far from contact (Dec = 3.6°, Inc = 12.9°, α_{95} = 20.4°). (b) Isothermal remanent magnetization (IRM) acquisition curves (normalized intensities versus magnetic field) for samples close and far from contact. Hysteresis curves showing wasp-waisted type characteristic of mixture phases: (c) close to contact and (d) far from contact.

the temperature contrast to the host rock. Simple thermal conduction models provide crude estimations on the time of cooling of igneous intrusions (Jaeger, 1968). It depends on the square of the half-tickness of the intrusion. For the Avanavero sills, the complete cooling of the intrusions considering their thickness would be of ~100 years for Paiuá, Camararém and Cotingo, and 5000 years for Pedra Preta. It is worth noting that each sill has intruded at different depths into the sedimentary succession, along more than 900 m of stratigraphic thickness implying different cooling rates and in consequence different remanence blocking times. From the 13 sites included in the Avanavero mean direction, seven are from the Pedra Preta sill, four are from the Cotingo sill, one is from the Camararem sill and one is from the Paiuá sill. The Pedra Preta sill, which concentrates most of the sampling sites, is the uppermost sill in the stratigraphy (see schematic profile in Fig. 1). Such a large intrusive rock would take at least thousands of years to cool. This is coherent with the medium to coarse grain sizes of the Avanavero rocks (diabases up to gabbros) as described above. Although the other sills are thinner they intruded different levels in the stratigraphy and likely record the field at different times. In this way, we believe site mean directions shown in Fig. 4 represent records of paleosecular variation which was averaged out in the general mean.

The Avanavero pole (AV: 27.9°E, 48.4°S, N = 13, A_{95} = 9.6°, K = 19.6) satisfies 5 out of the 7 quality criteria proposed by Van der Voo (1990) as follows: (1) U–Pb dating on dolerites from several sills and dykes yielded well-defined and concordant ages, with a mean of 1788.5 ± 2.5 Ma, which likely represent the age of magnetization acquisition; (2) the AV pole was derived from a set of 153 specimens (13 sites) and has adequate Fisher's statistical parameters (A_{95} = 9.6°, K = 19.8); (3) remanence vectors were isolated by least-squares fit from orthogonal diagrams (Kirschvink, 1980) after stepwise AF and thermal demagnetization; (5) the Avanavero rocks in the investigated area are situated in the Archean Amazonia-Central Province, which remained stable during the evolution of the Ventuari-Tapajós (1980–1810 Ma) and Rio Negro-Juruena (1780–1550 Ma) provinces, and the ca. 1200 Ma K'Mudku event seemed to have only slightly affected the Avanavero rocks at its southern border; (7) similar directions are not found in young units from the same region or cratonic block. Two criteria were not satisfied: (4) the baked contact test is not complete since we cannot prove that the direction found in the host rock far from the contact is primary or older than that of the intrusion; and (6) only a single polarity was found for these rocks. In sum, the Avanavero pole (AV) may be regarded as a robust pole (Q = 5), and as such it can be used to constrain the paleogeographic position of the proto-Amazonian Craton at ~1789 Ma and test the paleogeography of the proto-Amazonian Craton in the Columbia Supercontinent.

8.2. SAMBA connection and Laurentia in Columbia

A paleomagnetic based reconstruction of the Columbia Supercontinent at ~1780 Ma was presented by Bispo-Santos et al. (2008). In their reconstruction Laurentia, Baltica, North China and Amazonian Craton were laterally disposed along a Paleo- to Mesoproterozoic subduction-related mobile belt at the western side of the supercontinent (Fig. 10a). The position of the Amazonian Craton in that reconstruction was constrained by the 1780 Ma Colíder Group paleomagnetic pole. Recently, D'Agrella-Filho et al. (2012), interpreting new paleomagnetic data from the 1416 Ma Indiavaí mafic intrusive, proposed a very similar configuration (Fig. 10b), but which incorporated rotations within Baltica. In their model, Sarmatia (together with Amazonia and West Africa) were rotated 43° counterclockwise about an Euler pole at 48°N, 42.2°E relative to the Fennoscandia/Volgo-Uralia block following Bogdanova et al. (2008) and Elming et al. (2010). In their interpretation, D'Agrella-Filho et al. (2012) placed the North China block between Fennoscandia and Amazonian Craton, but suggested a triple junction between Fennoscandia/Sarmatia, North China and

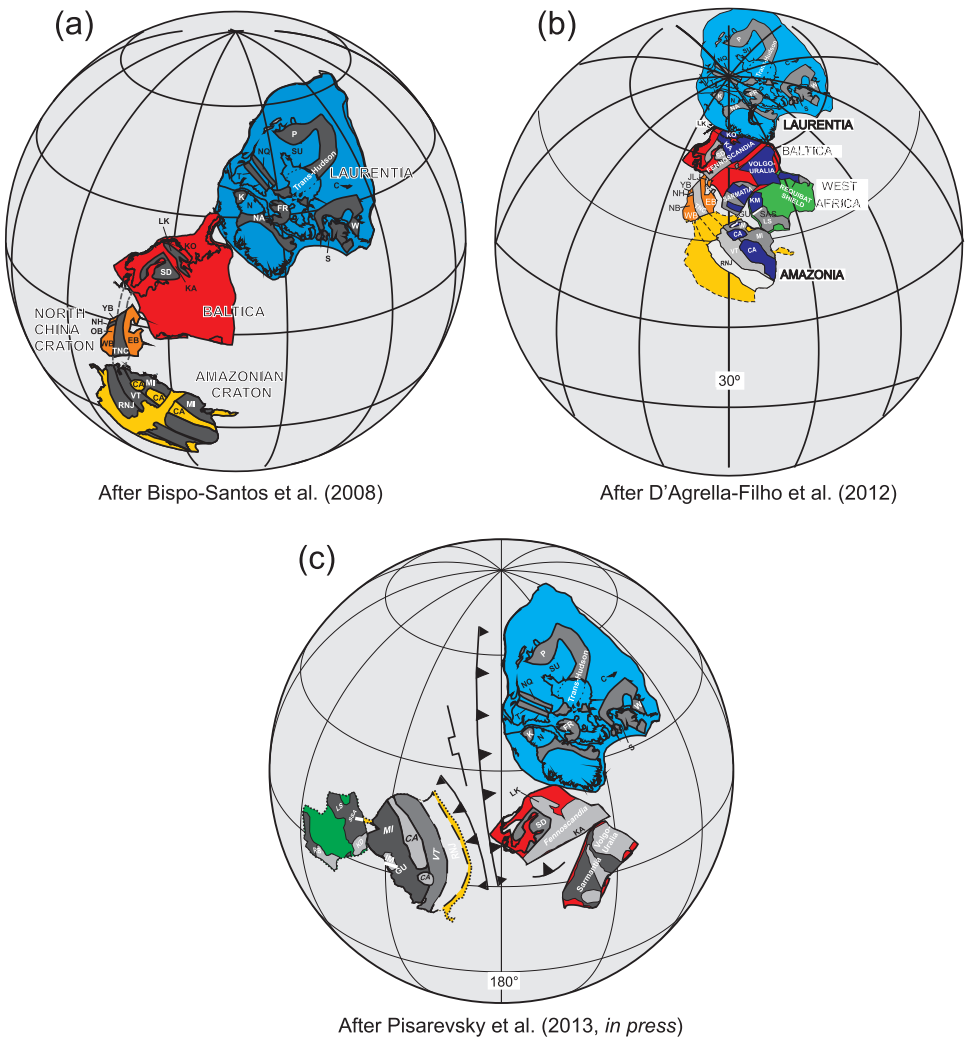


Fig. 10. Paleogeographic reconstruction at ~1780 Ma for Columbia Supercontinent, after (a) Bispo-Santos et al. (2008), (b) D'Agrella-Filho et al. (2012) and (c) Pisarevsky et al. (2014). Laurentia (LA) in blue; Baltica (B) in red; Amazonian Craton (AC) in yellow; North China Craton (NCC) in orange; and West Africa Craton (WA) in green. Archaean cratonic areas and Paleoproterozoic belts (dark gray): Laurentia (S – Slave; C – Churchill; SU – Superior; N – Nain, NQ – New Quebec; T – Torngat; W – Wopmay; P – Penokean; K – Kefilidian; NA – Nagssugtoqidian; FR – Foxe-Rinklan), Baltica (KO – Kola; KA – Karelia, LK – Lapland-Kola; SD – Svecofennian Domain; G – Gothian Province), Amazonia (CA – Central Amazonian, MI – Maroni-Itacaiunas; VT – Ventuari-Tapajos; RNJ – Rio Negro-Juruena, GU – Guri lineament) and West Africa Craton (LS – Leo Shield, KD – Kenemanan Domain. RB – Requibat Shield, SSA – Sassandra lineament) and North China Craton (YB – Yinshan block; OB – Ordos Block; WB – West Block; EB – East Block, NH – North Hebei/Khondolite belt; TNC – Trans-North China belt; JLJ – Jiao-Liao-Ji belt). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Amazonia; the break-up of North China and its drift away from the supercontinent occurring at 1780 Ma ago. According to those authors the profuse 1790–1780 Ma anorogenic intraplate magmatism in Fennoscandia, Amazonia and North China would be the record of this rupture. They include: the 1780 Ma dykes of the Småland Province in south Fennoscandia (Pisarevsky and Bylund, 2010), the 1789 Ma dykes and sills from the Avanavero LIP in northern Amazonian Craton, and the 1780 Ma dykes widely spread in north and south of North China (Kusky et al., 2007). Yet, even though the Baltica/North China/Amazonia link was supported by the 1780 Ma Colíder Group pole (Bispo-Santos et al., 2008; D'Agrella-Filho et al., 2012), most authors consider that Baltica was directly linked to proto-Amazonian Craton at that time, with no intervening block (Geraldes et al., 2001; Zhao et al., 2004, 2006; Hou et al., 2008; Johansson, 2009; among others). This model has received the name of SAMBA after Johansson (2009), in a paleogeography of Columbia where Laurentia, Baltica, Amazonia and West Africa Cratons formed a great continental mass, which remained together for 500 Ma, between 1800 and 1300 Ma.

The new 1789 Ma paleomagnetic pole obtained for the Avanavero sills allows us to test the SAMBA connection. Fig. 11 shows a reconstruction for Laurentia, Baltica, Amazonia and West Africa Cratons using the following Euler rotation poles: 16.8°N, 323.6°E; −64.6° for Laurentia, 6.3°N, 281.3°E, −41.1° for Baltica, 56.6°N, 152.9°E, 80.1° for proto-Amazonia, and 9.6°N, 182.7°E, 93.4° for West Africa Craton. In this reconstruction, correlatable geological provinces adjust according to the model of Johansson (2009). Also, the Guri (in Guiana Shield, Amazonian Craton) and Sassandra (in West Africa Craton) mega-shear zones are aligned as suggested by geological and paleomagnetic data (e.g., Onstott and Hargraves, 1981; Nomade et al., 2003; Evans and Mitchell, 2011; Bispo-Santos et al., 2014). Available paleomagnetic poles for Laurentia, Baltica and Amazonian Craton between 1800 Ma and 1750 Ma are presented in Table 2, and plotted in Fig. 11 after rotating them according to rotation poles assigned for each continent.

For Amazonia, besides the Avanavero pole represented by AC-1, Onstott et al. (1984) published paleomagnetic results (their component II) for the Aro and Guaniamo dykes from Venezuela, which

Table 2
Available paleomagnetic poles between 1800 Ma and 1750 Ma from Laurentia, Baltica and Amazonian Craton.

Continent/formation	Plat (°N)	Plong (°E)	A_{95} (°) dp/dm	Euler poles[a]	RPlat (°N)	RLong (°E)	Age (Ma)	Ref.
Laurentia				16.8°N; 323.6°E (−64.6°)				
Mean (LA-1)[b]	20.8	265.5	5.2		69.0	296.7	1760–1740	1
Baltica				6.3°N; 281.3°E (−41.1°)				
Smaland Intrusions (BA-1)	45.7	182.7	8.0		81.8	119.9	1784–1769	2
Ropruchey Sills (BA-2)	40.5	229.8	8.1		63.2	273.9	1751 ± 3	3, 4
Shoksha Formation (BA-3)	42.0	221.0	7.0		69.7	268.7	1790–1750	5
Hoting Gabbro (BA-4)	43.0	233.3	12.1		62.2	281.4	1786–1610	4, 6
Amazonian Craton				56.5°N; 152.9°E (80.1°)				
Avanavero mafic sills (AC-1)	48.4	207.9	9.6		90.0	22.4	1788.5 ± 2.5	7
Rio Aro-Guaniamo Dykes (AC-2)	42.0	180.0	6.0		69.0	195.6	1820	8
Colider volcanic Suite (AC-3)	−63.3	298.8	10.2				1789 ± 7	9
West Africa Craton				9.6°N; 182.7°E (93.4°)				7

Plat – pole latitude; Plong – pole longitude; A_{95}, dp, dm – Fisher's statistic parameters.
[a] Euler rotation poles used to rotate poles and continents to the Columbia configuration. RPlat/RPlong – rotated pole latitude/pole longitude.
[b] Mean pole was calculated from 19 pos-Hudsonian poles by Irving et al. (2004).
Ref.: 1 – Irving et al. (2004); 2 – Pisarevsky and Bylund (2010); 3 – Fedotova et al. (1999); 4 – Lubnina et al. (2012); 5 – Pisarevsky and Sokolov (2001); 6 – Elming et al. (2009); 7 – this study; 8 – Onstott et al. (1984); 9 – Bispo-Santos et al. (2008).

they combined with similar paleomagnetic results obtained by Hargraves (1968) and Onstott and Hargraves (1981) for dolerites from other parts of Venezuela and Guyana. A mean pole (42°N, 180°E, A_{95} = 6.0°) was calculated for these intrusions, whose age was suggested to be ca. 1820 ± 20 Ma by Onstott et al. (1984) (see Table 2). This pole (AC-2 in Fig. 11) together with the Avanavero pole (AC-1) may be used to trace a short APW path (between 1820 Ma and 1790 Ma) for the proto-Amazonian craton (Fig. 11).

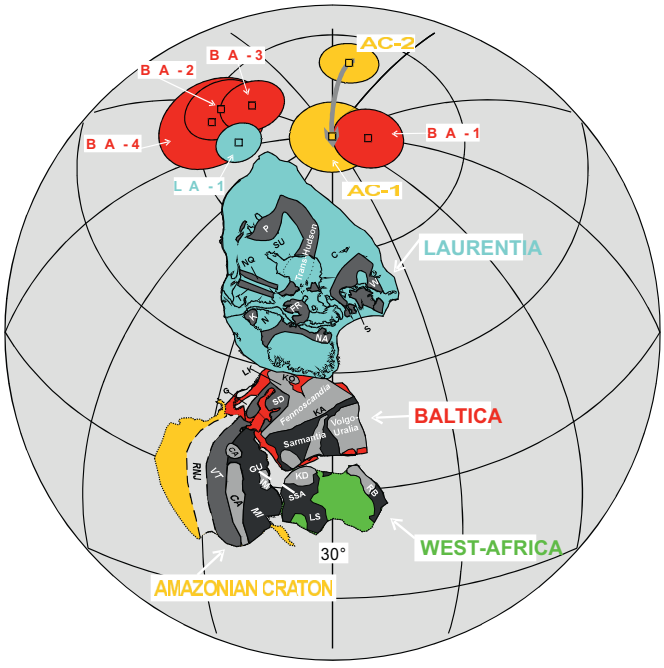


Fig. 11. Paleogeographic reconstruction for Laurentia, Baltica, proto-Amazonian Craton and West Africa at 1790–1780 Ma. Euler poles (Table 2) – Laurentia (16.8°N; 323.6°E; −64.6°), Baltica (6.3°N; 281.3°E; −41.1°), Amazonian Craton (56.5°N; 152.9°E; 80.1°) and West Africa (9.6°N; 182.7°E; 93.4°). Amazonia/Baltica link as the SAMBA model of Johansson (2009). Paleomagnetic poles between 1800 Ma and 1750 Ma for these continental blocks are also shown after rotating them using the same Euler poles used for each continent: Laurentia (LA) in blue; Baltica (B) in red; Amazonian Craton (AC) in yellow; and West Africa Craton (WA) in green. Archaean cratonic areas and Paleoproterozoic belts (dark gray): Laurentia (S – Slave; C – Churchill; SU – Superior; N – Nain, NQ – New Quebec; T – Torngat; W – Wopmay; P – Penokean; K – Kefilidian; NA – Nagssugtoqidian; FR – Foxe-Rinklan), Baltica (KO – Kola; KA – Karelia, LK – Lapland-Kola; SD – Svecofennian Domain; G – Gothian Province), Amazonia (CA – Central Amazonian, MI – Maroni-Itacaiunas; VT – Ventuari-Tapajos; RNJ – Rio Negro-Juruena, GU – Guri lineament) and West Africa Craton (LS – Leo Shield, KD – Kenemanan Domain. RB – Requibat Shield, SSA – Sassandra lineament). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Four paleomagnetic poles are presently available for Baltica in the time interval between 1790 Ma and 1750 Ma (Table 2): the Ropruchey sill (Fedotova et al., 1999), the Shoksha Formation (Pisarevsky and Sokolov, 2001), the Hoting Gabbro (Elming et al., 2009) and the Småland Intrusive (Pisarevsky and Bylund, 2010). A U–Pb zircon age of 1786 ± 10 Ma is attributed to the Hoting Gabbro but younger 1650–1610 Ma ^{40}Ar–^{39}Ar biotite ages for the same rocks bring some uncertainties concerning the age of the Hoting pole. Until recently the Ropruchey sill's pole was constrained by a U–Pb zircon age of 1770 ± 12 Ma (Fedotova et al., 1999) but a precise baddeleyite age of 1751 ± 3 Ma recently obtained for these rocks suggests a younger age for it (Lubnina et al., 2012). This is consistent with the fact that this pole differs by some degrees from the 1784–1769 Ma Småland Intrusions pole (Pisarevsky and Bylund, 2010). The age of the Shoksha pole is constrained between 1800 Ma using a glauconite K–Ar age for the underlying Petrozavodsk Formation and the 1751 ± 3 Ma age of the Ropruchey sill that cuts the Shoksha Formation. The agreement between the Shoksha (BA-3) and Hoting (BA-4) poles with the well-dated Ropruchey sill pole (BA-2) (Fig. 11, Table 2) strongly suggests these three poles represent the Baltica position at about 1750 Ma ago. These poles also agree (within error) with the Laurentian 1750 Ma pole, a mean of nineteen post-Hudsonian poles with ages between 1760 and 1740 Ma calculated by Irving and colleagues (2004) (LA-1 in Fig. 11 and Table 2). In the Baltica/Amazonian Craton reconstruction proposed here, the older 1784–1769 Ma Småland Intrusions pole (BA-1 in Fig. 11) matches within error the ca. 1790 Ma Avanavero pole (AC-1) corroborating the SAMBA connection of Johansson (2009).

A similar Laurentia/Baltica/Amazonia configuration was also recently proposed by Mertanen and Pesonen (2012) based on 1530 Ma paleomagnetic poles from Laurentia, Baltica and Amazonian cratons. Although paleomagnetic data for West Africa for this time interval is actually absent, geological evidence and older 2100–1970 Ma paleomagnetic data corroborate the Amazonia/West Africa link shown in Fig. 11 (Onstott et al., 1984; Nomade et al., 2003; Evans and Mitchell, 2011; Bispo-Santos et al., 2014). This great continental mass was probably the core of the Columbia Supercontinent (Salminen and Pesonen, 2007). Its western margin, formed by the contiguous Laurentia, Baltica and proto-Amazonian craton, was the scenario of coeval Andine-type, subduction-related accretionary mobile belts during Paleo- to Mesoproterozoic, which formed the Yavapai-Mazatzal-Labradorian (1800–1400 Ma) in

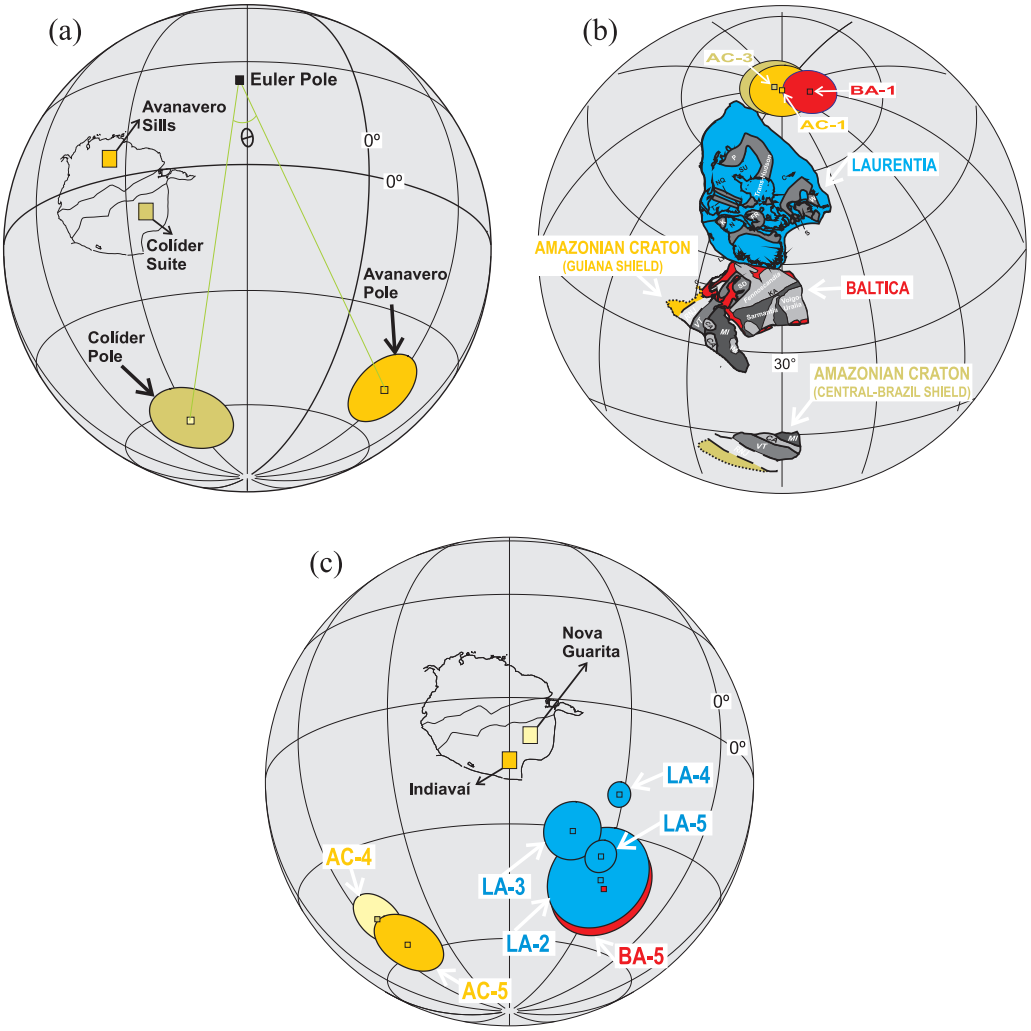


Fig. 12. (a) Comparison of the ca. 1790 Ma Avanavero and Colíder poles. Amazonian Craton in its present position (squares represent geographic locations of the Avanavero Sills and Colíder Group-investigated paleomagnetic areas). The Euler pole at 23.8°N, 327.6°E (47.7°) that takes the Colíder Group pole to match the Avanavero pole is also shown. Dextral strike-slip motions between these two areas after 1790 Ma could have been responsible for the present different positions of these two poles. (b) Relative positions of Guiana Shield (northern Amazonia) and Central-Brasil Shield (southern Amazonia) based on the 1790 Ma Avanavero and Colíder poles, respectively. In this reconstruction, we consider that both paleopoles result from primary directions. The Euler rotation pole (70.6°N, 64.1°E, 85.1°) was used to match Colider pole to Avanavero pole. (c) Comparison of the ca. 1460–1400 Ma poles (Table 3) from Amazonian Craton, Baltica and Laurentia. Amazonian Craton in its present position. Baltica and Laurentian poles were rotated using the following rotation Euler poles, respectively: 43.4°N, 200.3°E (−78.7°); 54.2°N, 249.2°E (−84.0°). These Euler poles bring Baltica and Laurentia to Amazonia (in its present position) corresponding to the same reconstruction presented in Fig. 11. Note that as in figure (a), the Nova Guarita and Indiavaí Mesoproterozoic poles (in yellow) from southeastern Amazonian Craton (geographic locations shown as square symbols) are located to the west of the Baltica and Laurentia poles (in red and blue, respectively), suggesting that if these movements occurred, most of them happened after ca. 1420 Ma ago. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Laurentia, the trans-Scandinavian-Kongsbergian (1800–1700 Ma) and Gothian (1660–1520 Ma) in Baltica, and the Ventuari-Tapajós (1950–1800 Ma) and Rio Negro-Juruena (1850–1550 Ma) in the Amazonian Craton (Hoffman, 1988; Bettencourt et al., 1996; Geraldes et al., 2001; Sadowski, 2002; Johansson, 2009; Mertanen and Pesonen, 2005, 2012).

Recently, Pisarevsky et al. (2014) rejected the SAMBA model and proposed that the Amazonia/West Africa never took part in Columbia. In their model (Fig. 10c), subduction-related zones in Amazonia/West Africa and Laurentia/Baltica correspond to opposite margins of an ocean and have opposite sense. The main reasoning for that is that Baltica was not yet completely assembled at 1790 Ma (Bogdanova et al., 2008; Elming et al., 2010). In a recent paper, Bogdanova et al. (2013) suggest an oblique collision of Sarmatia/Volgo-Uralia with Fennoscandia at 1800–1770 Ma, and a posterior counterclockwise rotation of Sarmatia/Volgo-Uralia to explain the 1800–1750 Ma mafic dyke intrusions related to an AMCG suite that are widespread in the Ukranian Shield. However, as stated by Bogdanova et al. (2013), 'the presently available paleomagnetic data for 1.75 Ga are still too limited to decide whether there was a wide oceanic gap between Fennoscandia and Volgo-Sarmatia at that time, or whether that gap was much narrower or even nonexistent'.

8.3. *Avanavero pole versus Colíder Group pole*

The Baltica/Amazonia link in the Columbia Supercontinent proposed in Fig. 11 contrasts with the model proposed by Bispo-Santos et al. (2008) and D'Agrella-Filho et al. (2012), based on the Colíder Group pole, where North China would be placed between Baltica and the Amazonian Craton.

The 1790 Ma Avanavero pole was obtained for anorogenic rocks emplaced in an intracratonic environment (Guyana Shield, northern Amazonian Craton), whose Archaean to Paleoproterozoic rocks were only partially affected by the ca. 1200 Ma K'Mudku event at its southern part (Cordani et al., 2010). As stressed above, the

Table 3
Available paleomagnetic poles between 1460 Ma and 1400 Ma from Laurentia, Baltica and Amazonian Craton.

Continent/formation	Plat (°N)	Plong (°E)	A_{95} (°) dp/dm	Euler poles[a]	RPlat (°N)	RLong (°E)	Age (Ma)	Ref.
Laurentia				54.2°N; 249.2°E (−84.0°)				
Michikamau Intrusives	2.0	38.0	3/7				1460 ± 5	1
Harp Lake Complex	−2.0	26.0	3/6				1450 ± 5	2
St. Francois Mountains	13.0	39.0	7.0				1476 ± 16	3
Tobacco Root Mountains	−9.0	36.0	10.0				1448 ± 49	4
Mean (LA-2)[#]	1.0	35.0	13.0		−41.8	329.9	1460	5
Mcnamara (LA-3)	14.0	28.0	7.0		−30.9	317.4	1401 ± 6	6
Electra Lake Gabbro (LA-4)	21.0	41.0	3.0		−21.0	328.7	1433 ± 2	7
Laramie Anorthosite (LA-5)	7.0	35.0	4.0		−36.1	327.5	1429 ± 9	8
Baltica				43.4°N; 200.3°E (−78.8°)				
Tuna Dykes	−21.0	0.0	7.0				1461–1462	9
Salmi Form. volc. rocks	−6.0	20.0	11.0				1460	9
Lake Ladoka Dolerites	−16.0	357.0	4.0				1457 ± 2	9
Valaam Monzodioritic – A pole	−14.0	346.0	2.0				1458 ± 4	9
Lake Ladoka mafic rocks	−15.0	357.0	6.0				1452 ± 12	10
Mean (BA-5)[#]	−15.0	0.0	13.0		−48.8	332.1	1460	5
Amazonian Craton								
Nova Guarita Dykes (AC-4)	−48.0	246.0	7.0				1414 ± 4	11
Indiavaí Intrusive (AC-5)	−57.0	250.0	9.0				1416 ± 7	12

Plat – pole latitude; Plong – pole longitude; A_{95}, dp, dm – Fisher's statistic parameters.
[a] Euler rotation poles used to rotate poles and continents to the Columbia configuration. RPlat/RPlong – rotated pole latitude/pole longitude.
[#] Mean for 1460 Ma.
Ref.: 1 – Emslie et al. (1976); 2 – Irving et al. (1977); 3 – Meert and Stuckey (2002); 4 – Harlan et al. (2008); 5 – this study; 6 – Elston et al. (2002); 7 – Harlan and Geissman (1998); 8 – Harlan et al. (1994); 9 – Salminen and Pesonen (2007); 10 – Lubnina et al. (2010); 11 – Bispo-Santos et al. (2012); 12 – D'Agrella-Filho et al. (2012).

Avanavero ChRM direction is likely a primary thermo-remanent component. Meanwhile the 1790 Ma Colíder Group pole was obtained for felsic rocks from southern Amazonian Craton, where NW-SE magmatic arcs (forming the Jauru Terrane) were in progress up to 1.32 Ga, with the final collision of the Paraguá Terrane (Bettencourt et al., 2010). No field tests were performed for the Colíder Group because of poor outcropping conditions. An easy interpretation is that the Colíder magnetization represents a secondary component, and it cannot constrain the position of the Amazonian Craton at 1790 Ma ago. In this case the SAMBA model of Johansson (2009) could have prevailed at 1790–1780 Ma as indicated by the Avanavero pole.

Alternatively, the Colíder Group pole could also represent a primary magnetization. Its stable, high-coercivity and high-unblocking temperature ChRM component seems to be carried by primary magnetite and/or hematite (Bispo-Santos et al., 2008). If both Avanavero and Colíder Group poles represent primary magnetizations acquired at about 1790 Ma, then relative tectonic movements must have occurred between the northern Avanavero area and the southern Colíder area after 1790 Ma. Fig. 12a shows the Avanavero and Colíder Group poles and the Amazonian Craton in its present position. The Colíder Group pole is located to the west of the Avanavero pole, and a rotation of 47.7° of the Colíder Group pole about an Euler pole at 23.8°N, 327.6°E (Fig. 12a), brings both poles together. Approximately NW dextral strike-slip motions between the northern Avanavero area relative to the southern Colíder area could explain the different positions of these poles (Fig. 12b).

The Mesoproterozoic (ca. 1420 Ma) Nova Guarita (Bispo-Santos et al., 2012) and Indiavaí (D'Agrella-Filho et al., 2012) poles from southern Amazonian Craton are also plotted to the west of the 1460–1420 Ma poles from Baltica and Laurentia in the SAMBA model (Johansson, 2009) (Table 3, Fig. 12c). This suggests that most of these movements, if they occurred, must have happened after 1420 Ma ago.

The recent recognition of the Trincheira ophiolite (see topic 2) at the southwestern edge of the Amazonian block suggests the docking of the Paraguá block to the proto-Amazonian craton along the Alto Guaporé Belt to have occurred at the Middle Mesoproterozoic, between ~1470 and 1320 Ma (Bettencourt et al., 2010; Rizzotto and Hartmann, 2012). NW-SE lineaments (the Buiuçu shear zone) are observed to the east of the Trincheira ophiolite, their mylonitic rocks being dated at 1466.5 ± 1.4 Ma (Ar–Ar in muscovite) and 1467.8 ± 0.8 Ma (Ar–Ar in sericite). These shear zones are interpreted as a result of the Rondonian-San Ignacio Orogeny (Almeida et al., 2012), which culminated with the Paraguá block collision. We speculate that if both paleomagnetic poles result from primary directions, reactivations of these faults can be, at least in part, responsible for the NW-SE dextral movements implied by the available paleomagnetic data.

9. Conclusions

This study presents paleomagnetic data performed on mafic sills from Avanavero magmatism, northern Roraima State (northern Amazonian Craton), with a mean age of 1788.5 ± 2.5 Ma (U–Pb, zircon). AF and thermal treatments revealed southeastern directions with downward/upward inclinations, carried by moderate to high-Hc and high-unblocking temperatures Ti-poor titanomagnetite/magnetite. Directions obtained from the host rocks close to the intrusion match the ones disclosed for the sills. Directions obtained in sandstones far from the intrusions are different. But we cannot claim for a positive baked contact test because the directions of the host sedimentary rocks away from the contact can result from a much younger remagnetization. The calculated mean direction (D_m = 138.2°; I_m = −3.4°; N = 13, α_{95} = 13.0°) for the Avanavero sills, yielded the paleomagnetic pole (AV pole) at 27.9°E, 48.4°S (A_{95} = 9.6°), which can be classified with a reliable factor Q = 5. The presently available Paleoproterozoic paleomagnetic poles corroborate the SAMBA model of Johansson (2009), with a nearly equatorial Columbia Supercontinent being formed by Laurentia, Baltica, proto-Amazonian Craton and West Africa, although other configurations are also possible (e.g., Pisarevsky et al., 2014).

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